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Fig. 5: Lethal footprint of HPM weapons

with a particular design depending upon the intended application.

Capabilities and limitations of HPM weaponisation

A number of technological and operational issues need to be addressed if HPMs were to move out of the confines of research and application laboratories to become serious contenders as operational weapons for defensive and offensive roles. These include issues of weapon systems' compactness and efficiency to suit a range of military platforms, issues related to antenna aperture to get desired operational ranges and peak-versus-average powers. Target acquisition, tracking and beam-pointing technologies and lethality assessment are other important factors.

Key advantages of HPM based directed energy weapons include speed-of-light delivery, all-weather capability, effectiveness over large areas, scalable effects, deep magazines and minimised collateral damage.

HPMs are speed-of-light weapons with essentially no time-of-flight as compared to finite travel time of

conventional kinetic energy weapons. A supersonic missile travelling at 6Mach speed would take a flight time of about 50 seconds to hit a target 300km away. On the other hand, an HPM weapon could hit the same target in a millisecond or so. This feature is particularly suitable for attacking fast-moving targets.

Unlike high-energy laser weapons (to be described later), which are clear-weather weapons, HPM weapons are not adversely affected by weather and atmospheric conditions. For all practical purposes, these are all-weather weapons.

HPM weapons are area weapons unlike high-energy laser weapons that are point weapons. The footprint depends on the weapon's power output, distance from the target and beam divergence. These affect all vulnerable equipment that comes within their lethal footprint (Fig. 5).

HPM weapons are ideally suited for targeting in urban environment due to minimised collateral damage. Some collateral damage may occur if friendly or civilian electrical systems are also in the footprint. HPM weapons have a deep magazine. Unlike conventional kinetic